# Linux Master Cheatsheet

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1) BASIC COMMANDS

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ls → List files
ls -la → List all with permissions
cd /path → Change directory
pwd → Show current directory
mkdir folder → Create folder
rm file → Remove file
rm -r folder → Remove folder recursively
cp src dst → Copy file
mv src dst → Move/rename file
cat file → Show file content
nano file → Simple text editor
less file → View file (scrollable)

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2) PERMISSIONS & OWNERSHIP

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chmod 755 file → rwxr-xr-x
chmod 644 file → rw-r--r--
chown user file → Change owner
chown user:group file → Change owner and group

Permission bits:
- r = read
- w = write
- x = execute

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3) SYSTEM MONITORING

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top → Real-time process viewer
htop → Better top (if installed)
ps aux → List all processes
free -h → RAM usage
df -h → Disk usage

du -sh *                →  Size of folders in current dir

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4) SERVICES (systemd)

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systemctl status service      →  Show service status
systemctl start service       →  Start service
systemctl stop service        →  Stop service
systemctl restart service     →  Restart service
systemctl enable service      →  Enable at boot
systemctl disable service     →  Disable at boot

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5) NETWORKING

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ip a                    →  Show network interfaces
ip r                    →  Show routes
ping 8.8.8.8            →  Test connectivity
ping google.com         →  Test DNS + connectivity
ss -tulnp               →  Show open ports
curl http://example.com  →  Test HTTP
wget URL                →  Download file

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6) SSH BASICS

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ssh user@host           →  Connect to remote machine
ssh -p 2222 user@host   →  SSH on custom port
scp file user@host:/path →  Copy file to remote host
ssh-keygen              →  Generate SSH key pair
ssh-copy-id user@host   →  Copy public key to server

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7) DISKS & PARTITIONS

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lsblk                   →  List block devices (disks)
blkid                   →  Show UUIDs of partitions
df -h                   →  Show mounted filesystem usage
mount /dev/sdXN /mnt    →  Mount partition
umount /mnt             →  Unmount partition
fdisk -l                →  List disks and partitions (MBR)
gdisk -l /dev/sdX       →  GPT-aware partition tool

Common layout:
- /dev/sda → Disk
- /dev/sda1, /dev/sda2 → Partitions

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8) FILESYSTEM STRUCTURE (OVERVIEW)
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/ → Root of filesystem
/home → User home directories
/etc → Configuration files
/var → Logs, variable data
/tmp → Temporary files
/boot → Kernel and bootloader files
/usr → User applications and libraries

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9) LINUX BOOT PROCESS (UEFI + GRUB)
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1) Firmware (UEFI or BIOS) starts.
2) In UEFI mode, firmware reads entries from NVRAM.
3) UEFI loads the bootloader from the EFI partition, e.g.:
   /EFI/ubuntu/grubx64.efi
   /EFI/Boot/bootx64.efi (fallback)
4) GRUB is displayed (menu).
5) GRUB loads the Linux kernel (vmlinuz).
6) Kernel loads initramfs and mounts root filesystem.
7) systemd (PID 1) starts services and targets.

When dual-booting with Windows:
- Linux and Windows each have their own directory under /EFI.
- GRUB can chainload Windows Boot Manager.

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10) SHORTCUTS IN TERMINAL
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Ctrl + C → Cancel running command
Ctrl + D → End of input / logout
Ctrl + L → Clear screen
Ctrl + A → Go to beginning of line
Ctrl + E → Go to end of line
Ctrl + R → Search command history

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11) TROUBLESHOOTING TIPS

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"Permission denied":

- Use: sudo command
- Or fix permissions with chmod / chown carefully.

Port already in use:

ss -tulnp | grep 3000

kill -9 PID

Disk not showing:

lsblk

dmesg | tail

Low space:

df -h

du -sh *

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12) SECURITY & HARDENING (BASICS)

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- Use sudo instead of logging in as root.
- Keep packages updated:
  sudo apt update && sudo apt upgrade
- Restrict SSH:
  - Disable root login in /etc/ssh/sshd_config
  - Use SSH keys instead of passwords
- Restrict file permissions for sensitive files:
  chmod 600 ~/.ssh/id_rsa

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13) TEMP & LOGS

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/tmp → Temporary files

/var/tmp → Temporary but longer-lived

/var/log → System logs (syslog, auth.log, etc.)

You can view logs using:

  journalctl

  tail -f /var/log/syslog